



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Mid-Autumn Semester Examination 2024-25

Date of Examination: 19th September 2024 Session: AN Duration: 2 Hrs Full Marks: 50

Subject No. : AI61004 Subject : Statistical Foundations of AI/ML Department: Artificial Intelligence

Specific charts, graph paper, log book etc., required _____

Special Instructions (if any) : ALL CALCULATIONS NEED TO BE SHOWN IN FULL DETAILS

Please answer Q1 and any 3 of the remaining Questions. All parts of each question must be answered together.

Q1. A) Define unbiased and consistent estimators. Explain with a synthetic example (no need to consider any actual estimator) how an unbiased estimator need not be consistent and vice versa. **[4+2 = 6 marks]**

B) What is the Maximum Likelihood Estimator of the Gaussian mean 'u' based on 'n' IID observations following $N(u, s^2)$, where 's' is fixed and known? Is it unbiased and consistent? **[2+4= 6 marks]**

C) Give an example of an estimator that is biased, but asymptotically unbiased. Justify with full calculations. **[2+6 = 8 marks]**

Q2. A) Consider two independent 2D random variables $X \sim N(u_1, \Sigma_1)$ and $Y \sim N(u_2, \Sigma_2)$. Calculate the covariance matrix of $aX + bY$ where a, b are any real numbers. **[5 marks]**

B) Calculate the mean square error of the maximum likelihood estimator of the Bernoulli and Gaussian mean parameters, based on N IID observations? **[5 marks]**

Q3. A) We have a model $Y \sim N(W^T X, \sigma^2)$ where W, X are real vectors. Further, $W = V \cdot U$, i.e. W is obtained by elementwise multiplication between a real-valued vector V and a random

binary vector U such that each element of U follows Bernoulli distribution with parameter p . Write down the likelihood function. [3 marks]

B) We have an observation ($X = [2 \ 2]$, $Y=6$) from the above model. Assuming ($V = [3 \ 3]$, $\sigma^2 = 1$), what is the maximum likelihood estimate of p ? [3 marks]

C) $X_1 \sim \text{Gamma}(a, 1)$, $X_2 \sim \text{Gamma}(b, 1)$. Find the distribution of $Y = X_1/(X_1+X_2)$. [4 marks]

Q4. A) Show that Dirichlet distribution is a Conjugate Prior for Multinomial Distribution. [4 marks]

B) Consider the following model: $Z_i \sim \text{Beta}(a, b)$, $X_i \sim N(WZ_i, s)$. We have 3 observations of X and Z : [$X_1=4.6 \ Z_1=0.2$, $X_2=-2.4 \ Z_2=0.6$, $X_3=1.8 \ Z_3=0.4$]

Consider the following 2 choices of parameters: P1: [$W=5$, $s=9$, $a=4$, $b=4$], P2: [$W=3$, $s=25$, $a=2$, $b=2$]. Which of these two sets of parameters fits the observations better? [6 marks]

[$\text{Beta}(2, 2) = 1/6$, $\text{Beta}(4, 4) = 1/140$]

Q5. A) Derive the probabilistic version of ridge regression by considering $N(0, p^2 I)$ as the prior on the coefficient vector W . [4 marks]

B) We have 10 data-points of the form $D=\{x_i, y_i\}$ where (x_i, y_i) are both real numbers (given below). I want to fit a model $Y = WX + E$ where W is a real number and E is a Gaussian noise $N(0, 9)$. I have a prior distribution $p(W) = N(5, 1)$ on the parameter W . Derive the expected value for y_{11} , based on x_{11} and the first 10 observations (D). [6 marks]

ID	1	2	3	4	5	6	7	8	9	10	11
X	-4	-3	-2	-1	0	1	2	3	4	5	6
Y	-18	-8	-9	0	-3	5	5	11	20	19	?